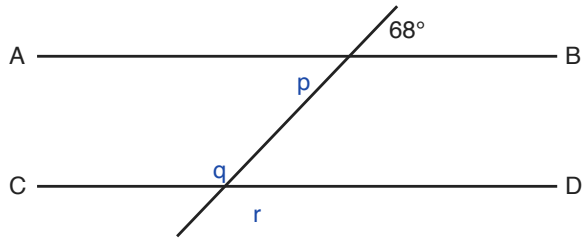
Shape and Space 1 — Angles · Triangles · Polygons · Similar triangles

Give a *reason* for every angle you find (e.g. "angles on a straight line", "alternate angles"). Diagrams are not drawn to scale. Marks in brackets. ★ = harder question.

A · ANGLES AND TRIANGLES

1 Lines AB and CD are parallel. A straight line crosses them as shown. Work out the angles p, q and r, giving a reason for each. [3]



2 In triangle PQR, $PQ = PR$. Angle $Q = (2x + 5)°$ and angle $P = (x + 10)°$. Find x and the size of each angle. [3]

3 In triangle ABC, the side BC is extended to a point D. Angle $ACD = 125°$ and angle $BAC = 70°$. Work out angle ABC, with a reason. [2]

B · QUADRILATERALS AND POLYGONS

4 The angles of a quadrilateral are $3x°$, $(2x + 20)°$, $(x + 40)°$ and $90°$. Find x and the four angles. [3]

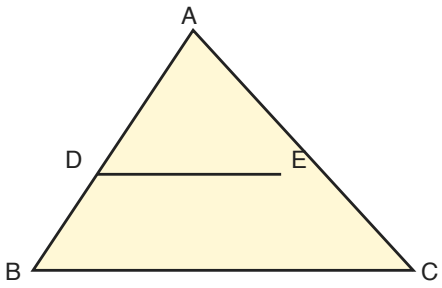
5 **a** Work out the interior angle of a regular decagon (10 sides). [4]

b A regular polygon has exterior angles of $24°$. How many sides does it have, and what is its interior angle?

6 A regular hexagon and a square share a side. At one end of the shared side, the hexagon and the square each have another side meeting at that vertex. Calculate the angle between those two sides, outside both shapes. Show your working. [3]

C · SIMILAR TRIANGLES AND SCALE

7 In triangle ABC, D lies on AB and E lies on AC with DE parallel to BC. $AD = 6$ cm, $DB = 4$ cm, $DE = 9$ cm and $AE = 7.5$ cm. [4]



a Explain why triangles ADE and ABC are similar.

b Work out the length BC.

c Work out the length EC.

8 A map has scale $1 : 50\,000$. [3]

a Two villages are 7.4 cm apart on the map. How far apart are they in real life, in km?

b A road is 12 km long. How long is it on the map?

9 Describe, step by step, how to construct with compasses and a ruler [2]

a the perpendicular bisector of a line segment AB of length 6 cm;

b an angle of $60°$.

10 At the same moment, a 1.8 m tall person casts a shadow 2.4 m long and a tree casts a shadow 14 m long. Use similar triangles to find the height of the tree. [3]

Total: 30 marks

Reasons to quote: angles on a line 180° · around a point 360° · vertically opposite · alternate / corresponding / co-interior (parallel lines) · triangle 180° · quadrilateral 360° · exterior angles of a polygon 360°.